

Architecture Justification

Domain & SSL Certificate Expiry Alert System

Architectural Choice :

Selected Architecture: Layered Architecture

We selected **Layered Architecture** for the Domain & SSL Certificate Expiry Alert System. The system is divided into separate components responsible for scheduling monitoring tasks, performing SSL/TLS and WHOIS checks, managing alerts, sending notifications, maintaining monitoring data, and displaying compliance information.

Reason 1 — Separation of Concerns :

Layered Architecture separates different responsibilities of the system into independent components. The **Monitoring Scheduler**, **SSL/TLS Monitor**, and **Domain/WHOIS Monitor** handle monitoring activities, while the **Alert & Escalation Manager** handles expiry alerts and escalation. The **Notification Service**, **Monitoring Database**, and **Compliance Dashboard** handle communication, data storage, and reporting respectively. This separation makes the system easier to understand, maintain, test, and modify.

Reason 2 — Maintainability and Scalability :

The architecture allows individual components to be modified or extended without significantly affecting the rest of the system. For example, a new notification channel can be added to the Notification Service, or additional monitoring functionality can be introduced without redesigning the complete system. This makes the architecture suitable for future expansion of the monitoring system.

Security Advantage :

Layered Architecture improves security by controlling how components communicate with each other. Users interact with the Compliance Dashboard rather than directly accessing the Monitoring Database or monitoring services. This separation helps protect monitoring information and limits direct access to sensitive system components.

Performance Benefit :

The architecture allows scheduled monitoring operations to run independently from dashboard access. The **Monitoring Scheduler** can initiate SSL/TLS and WHOIS checks while the **Compliance Dashboard** retrieves already-stored information from the database. This reduces unnecessary repeated external checks and allows users to view compliance information without waiting for a new monitoring operation.

Component Diagram:

